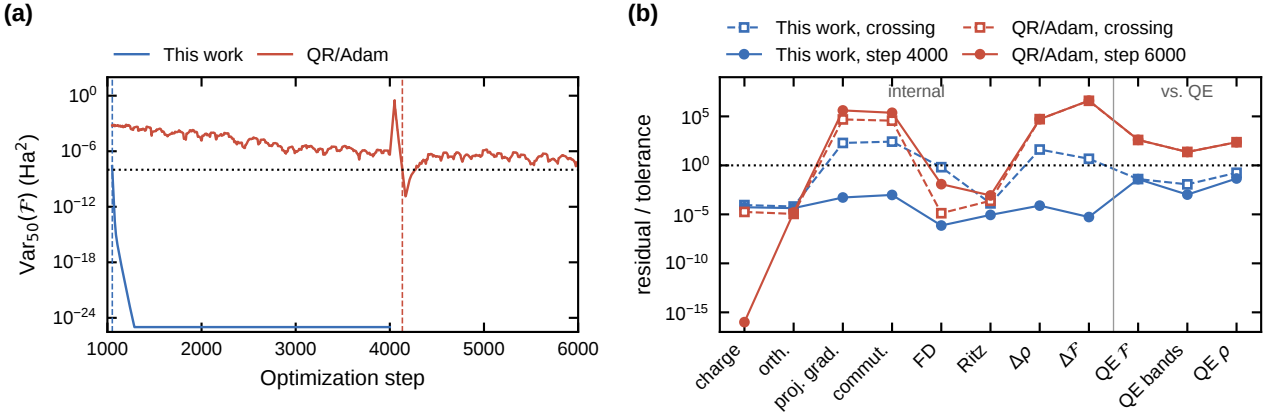
(a)
This work
QR/Adam
10^{0}
10^{-6}
10^{-12}
10^{-18}
10^{-24}
$\mathrm{Var}_{50}(\mathcal{F})$ (Ha^2)
1000
2000
3000
4000
5000
6000
Optimization step
(b)
This work, crossing
QR/Adam, crossing
This work, step 4000
QR/Adam, step 6000
Internal
vs. QE
10^{5}
10^{0}
10^{-5}
10^{-10}
10^{-15}
residual / tolerance
charge
orth.
proj. grad.
commut.
FD
Ritz
$\Delta\rho$
$\Delta\mathcal{F}$
QE $\mathcal{F}$
QE bands
QE ρ